Sean Luka Girgis

Python / AWS Software Engineer | Data Engineering, GenAI/RAG & Observability

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Professional Summary

Senior technology professional with 20+ years of enterprise experience across Python, SQL, AWS data platforms, REST/API concepts, data engineering, GenAI/RAG project work, Streamlit dashboards, observability, performance troubleshooting, and regulated financial-services delivery. Strong background building reliable Python data workflows, AI-ready datasets, dashboards, validation logic, operational reporting, and telemetry-driven systems in enterprise environments. Best aligned to Python/AWS roles requiring data engineering, GenAI/RAG enablement, API-oriented delivery, documentation, observability, CI/CD concepts, and stakeholder collaboration; FastAPI, ECS Fargate, API Gateway, Terraform/CloudFormation, OAuth/JWT, Bedrock SDKs, and production microservices are positioned carefully as active-learning or adjacent areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable extraction, transformation, querying, analytics, and downstream consumption.
- Developed Python-driven data workflows, validation logic, SQL transformations, dashboards, and reusable operational reporting patterns for regulated financial-services stakeholders.
- Prepared AI/ML-ready datasets for Prophet and scikit-learn forecasting workflows, supporting capacity planning, bottleneck prediction, trend analysis, and executive decision support.
- Built Streamlit dashboards and Python visualization workflows using matplotlib, seaborn, and plotly to surface capacity, performance, and reliability insights for technical and business teams.
- Supported observability and operational reliability by integrating telemetry from BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers.
- Troubleshot production data and monitoring issues by tracing source feeds, transformation logic, SQL behavior, metric behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Documented data flows, API/data-access assumptions, validation checkpoints, metric definitions, runbooks, support steps, and design notes to improve repeatability and cross-team knowledge sharing.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, supporting application performance visibility and operational readiness.
- Led the FAST project, mining real-user and synthetic monitoring metrics to surface optimization opportunities adopted by engineering teams.

- Built dashboards and reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and performance reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual performance-reporting workflows.
- Designed custom Management Modules, dashboards, alerts, SLA thresholds, and documentation standards used by application, middleware, infrastructure, and operations teams.
- Provided technical training, onboarding support, troubleshooting guidance, and documentation to improve adoption of monitoring standards and operational playbooks.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, regression thresholds, test findings, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Supported performance-focused cutover planning, validation, and troubleshooting for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java

Flagship Projects

AI-Powered Job Search Pipeline

Built a Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking.

Emphasized structured data processing, LLM API integration, validation, and reliable workflow execution.

Technologies: Python, JSON, REST/API concepts, FAISS, Sentence Transformers, Grok / xAI API, Quality Gates

ServiceCall AI RAG Demo

Built staged RAG proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, retrieval decisions, answer contract schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.

Technologies: Python, FastAPI, Docker, TF-IDF, Hybrid Retrieval, Pydantic, RAG

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for ingestion, transformation, curated analytics datasets, query optimization, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — Capacity Forecasting Engine

Built Python/PySpark forecasting workflows to process infrastructure telemetry, prepare forecasting datasets, validate pipeline outputs, and deliver capacity and bottleneck insights through dashboards.

Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, BMC TrueSight, Telemetry